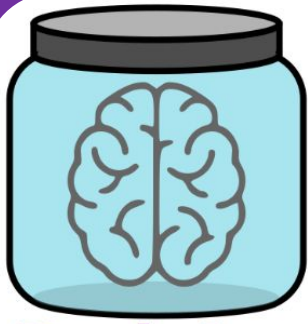




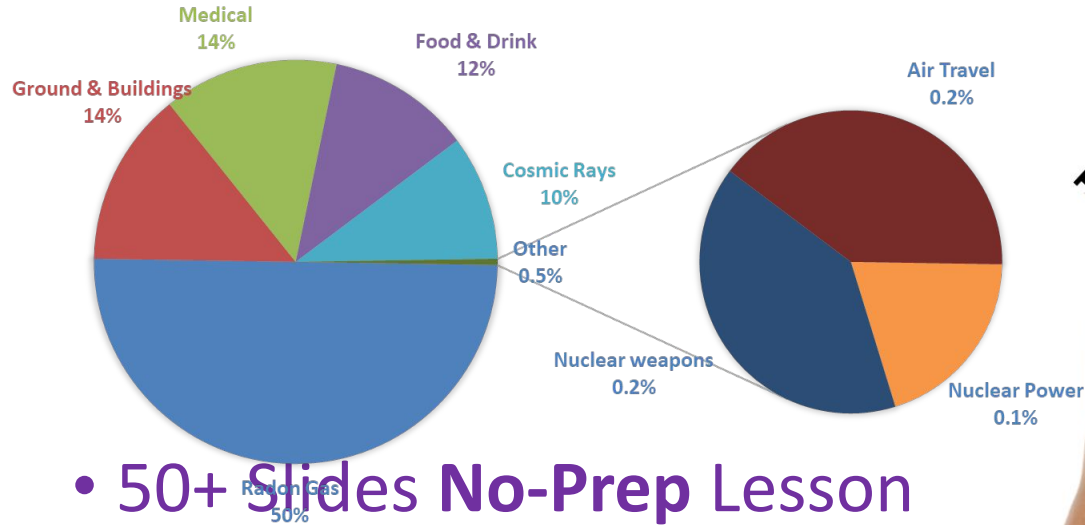
BrainJar
Teaching Resources

No-Prep Complete Lesson

GCSE Physics

50+ Slides

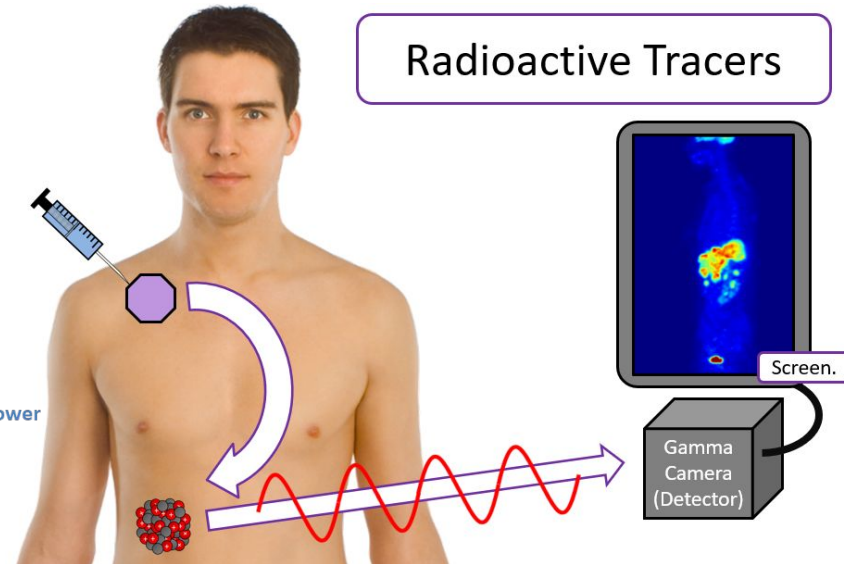
Radiation in Medicine & Background Radiation

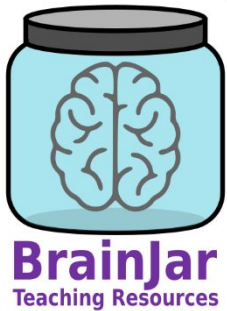


- 50+ Slides **No-Prep** Lesson

- Ideal for non-Physicists
- Animated Diagrams.
- Exam Style Questions
- **Answers**

Radioactive Tracers



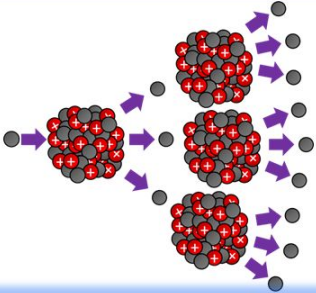


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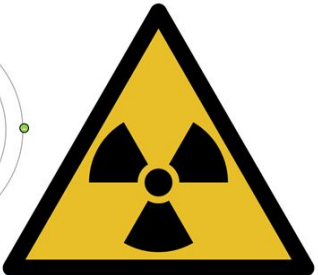
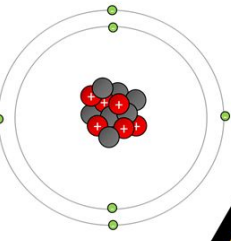
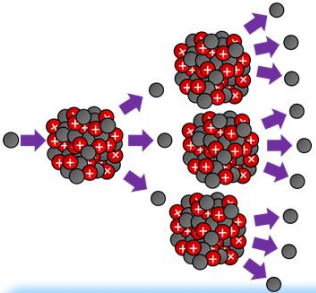
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GCSE Physics
Nuclear Fission
65+ Slides



- 65+ Slides **No-Prep Lesson**
- **Exam Style Questions**
- Ideal for non-Physicists
- **Answers**
- Anim

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Mega Bundle: Atomic Structure
AQA 4.4 / 6.4



- **No-Prep Lessons**
- **Exam Style Questions**
- Ideal for non-Physicists
- **Answers**
- Anima

Discounted Bundle

4.4.3 Hazards and uses of radioactive emissions and of background radiation (physics only)

4.4.3.2 Different half-lives of radioactive isotopes

Content

Radioactive isotopes have a very wide range of half-life values.

Students should be able to explain why the hazards associated with radioactive material differ according to the half-life involved.

4.4.3.3 Uses of nuclear radiation

Content

Nuclear radiations are used in medicine for the:

- exploration of internal organs
- control or destruction of unwanted tissue.

Students should be able to:

- describe and evaluate the uses of nuclear radiations for exploration of internal organs, and for control or destruction of unwanted tissue
- evaluate the perceived risks of using nuclear radiations in relation to given data and consequences.

4.4.3.1 Background radiation

Content

Background radiation is around us all of the time. It comes from:

- natural sources such as rocks and cosmic rays from space
- man-made sources such as the fallout from nuclear weapons testing and nuclear accidents.

The level of background radiation and radiation dose may be affected by occupation and/or location.

Radiation dose is measured in sieverts (Sv)

1000 millisieverts (mSv) = 1 sievert (Sv)

Students will not need to recall the unit of radiation dose.

Key: Items in a green box are covered in this lesson.

Teacher Notes

This lesson covers all of AQA GCSE Physics 4.4.3 (i.e. 4.4.3.1, 4.4.3.2, 4.4.3.3) which are required for GCSE Physics only and not combined science.

4.4.3.2 – explaining why the hazards of radioactive materials differ according to the half life – is covered through teaching about radiation in medicine and is also covered in subsequent lessons e.g. nuclear fission.

The section on sources of background radiation (which goes into detail about radon gas, cosmic rays) is to illustrate and help students understand sources of background radiation – in depth details are not needed.

This Institute of Physics worksheet activity is recommended as an engaging extension activity [Background Radiation: measuring your annual dose \(iop.org\)](#) and [CPSep08 2pp insert v6.indd \(iop.org\)](#)

Radiation in Medicine & Background Radiation

Starter:

1. Define “contamination”
2. Define “irradiation”
3. Which is more dangerous – a contaminated item or an irradiated item?
4. What two factors would you need to consider if you were deciding which radioactive source to use in a smoke detector?
5. Are alpha emitters more dangerous inside our outside the body? Explain your answer.

Answers

1. Define “contamination”: **The unwanted presence of materials containing radioactive isotopes on other materials.**
2. Define “irradiation”: **Irradiation is the process of exposing an object to nuclear radiation.**
3. Which is more dangerous – a contaminated item or an irradiated item? **A contaminated item as it will be radioactive, whereas an irradiated item is not radioactive.**
4. What two factors would you need to consider if you were deciding which radioactive source to use in a smoke detector? **The type of nuclear radiation emitted (i.e. alpha, beta, gamma) *and* the half life of the source.**
5. Are alpha emitters more dangerous inside our outside the body? Explain your answer. **Alpha emitters are more dangerous if they are inside out bodies. Alpha is the most ionising, but because it is not very penetrating it can be stopped by skin and therefore cannot damage your internal organs if the radioactive source is outside your body. However, if you were to swallow or breathe in a source of alpha radiation then the alpha radiation would not escape your body, and because it is highly ionising it would do a lot of damage to your cells.**

Lesson Focus and Learning Objectives

Lesson Focus:

How is radiation used in medicine?

What is background radiation?

Learning Objectives (To Be Able To...):

- Describe how nuclear radiation is used in medicine to explore internal organs and to control or destroy unwanted tissue.
- Explain why a specific radioactive source is most suitable for a given situation based on the type of radiation it emits and its half life.
- Explain why the hazards associated with a radioactive source is related to its half life.
- Describe what is meant by “background radiation” and its natural and artificial sources.
- Describe how we measure radiation dose.
- Explain why factors such as location and occupation can affect background radiation and radiation dose.

4.4.3 Hazards and uses of radioactive emissions and of background radiation (physics only)

4.4.3.2 Different half-lives of radioactive isotopes

Content

Radioactive isotopes have a very wide range of half-life values.

Students should be able to explain why the hazards associated with radioactive material differ according to the half-life involved.

4.4.3.3 Uses of nuclear radiation

Content

Nuclear radiations are used in medicine for the:

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- control or destruction of unwanted tissue.

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- evaluate the perceived risks of using nuclear radiations in relation to given data and consequences.

4.4.3.1 Background radiation

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Background radiation is around us all of the time. It comes from:

- natural sources such as rocks and cosmic rays from space
- man-made sources such as the fallout from nuclear weapons testing and nuclear accidents.

The level of background radiation and radiation dose may be affected by occupation and/or location.

Radiation dose is measured in sieverts (Sv)

1000 millisieverts (mSv) = 1 sievert (Sv)

Students will not need to recall the unit of radiation dose.

This is what we will cover in this lesson.

Using Radiation

Last lesson we looked at:

- The dangers of radiation.
- How we can use radioactive sources safely.
- Contamination.
- Irradiation.

We looked at some examples of how radiation can be used:

- Smoke alarms
- Automatic thickness monitoring.
- Checking for leaks in pipes.
- Irradiation to sterilise medical equipment and food.

We will build on this knowledge in this lesson.

Radiation in Medicine

Radiation can be used in medicine both for **diagnosis** and **treatment**.

There are two main ways that radiation is used in medicine:

- **Diagnosis:** exploration of internal organs.
- **Treatments:** control or destruction of unwanted tissue e.g. cancer cells.

Radioactive Tracers

Last lesson we looked at how a **radioactive tracer** could be used to find a leak in an underground pipe.

A similar process can be used to see what is happening inside your body.

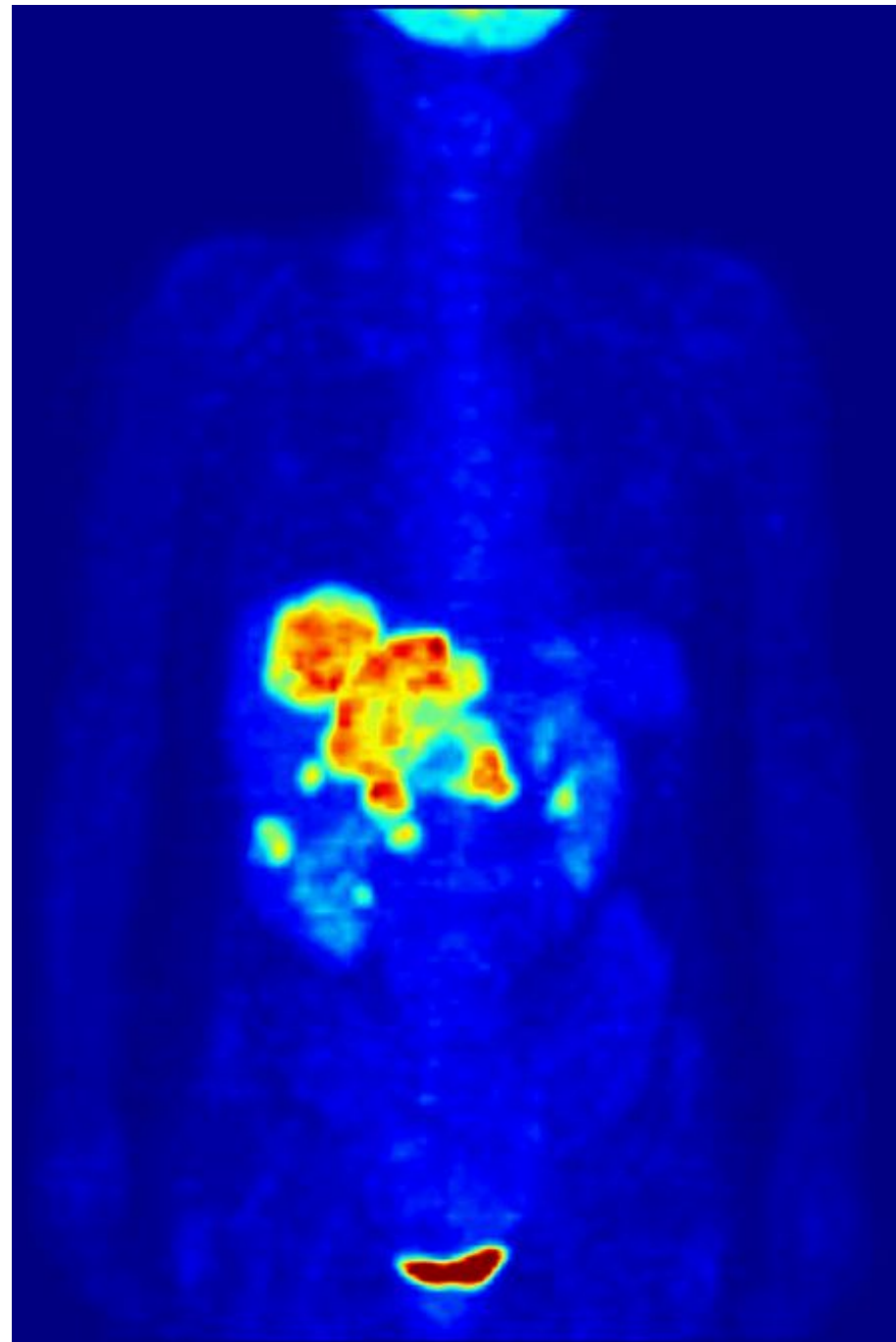
Radioactive Tracers

A radioactive source is swallowed by or injected into the patient. This is called the **radioactive tracer**.

The tracer contains a **radioactive isotope** that decays and emits gamma radiation.

The gamma rays can be **detected** outside of the patients body.

The radioactive isotope may be a single atom that makes up part of a compound. The particular compound may be chosen so that it concentrates in the specific part of the body we want to look at.

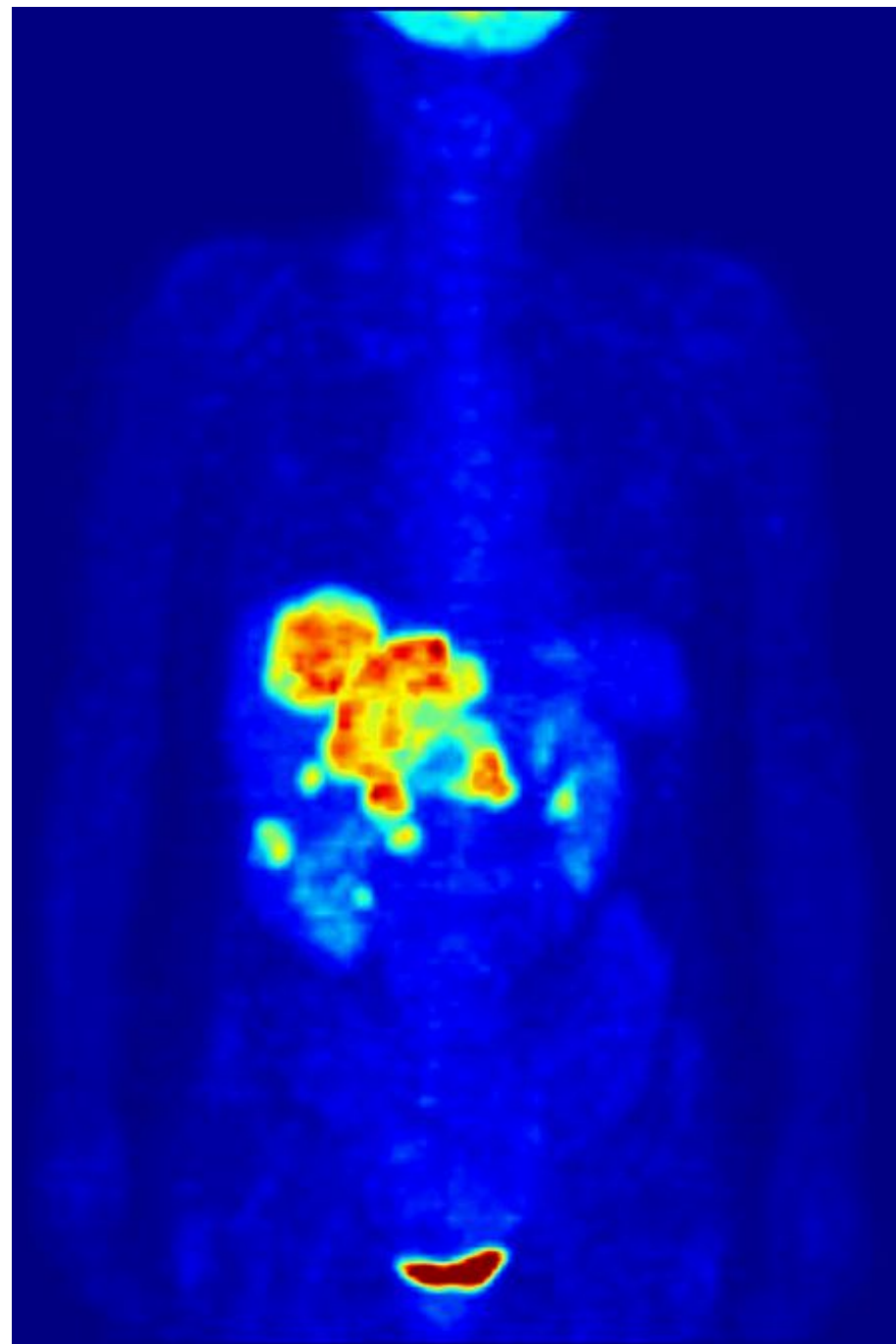


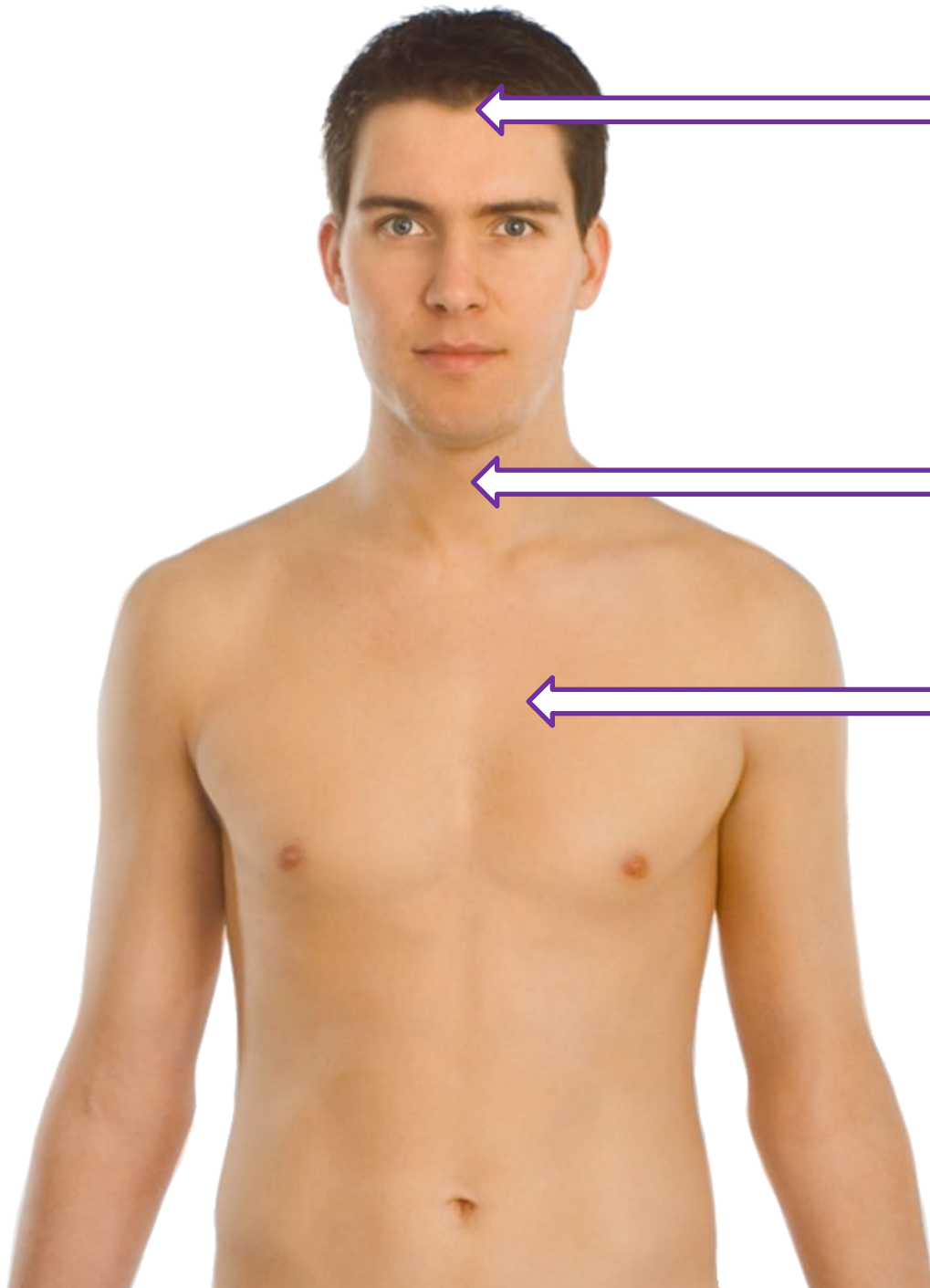
More Detail

A technique called Positron Emission Tomography (PET) has been used to make this particular image.

The patient was injected with a tracer called F-FDG, a compound that contains the radioactive isotope ^{18}F . F-FDG is similar to glucose and so concentrates where cells take up a lot of glucose. It is used to look for cancer cells which replicate rapidly and so use a lot of glucose.

The tracer has accumulated in the bladder, kidneys and brain, but the image also shows that the tracer is accumulating in cancerous tumours in the abdomen.





Brain: technetium-99

Thyroid gland: iodine-123

Heart: nitrogen-13

Different radioactive tracers can be used to study specific parts of the body.

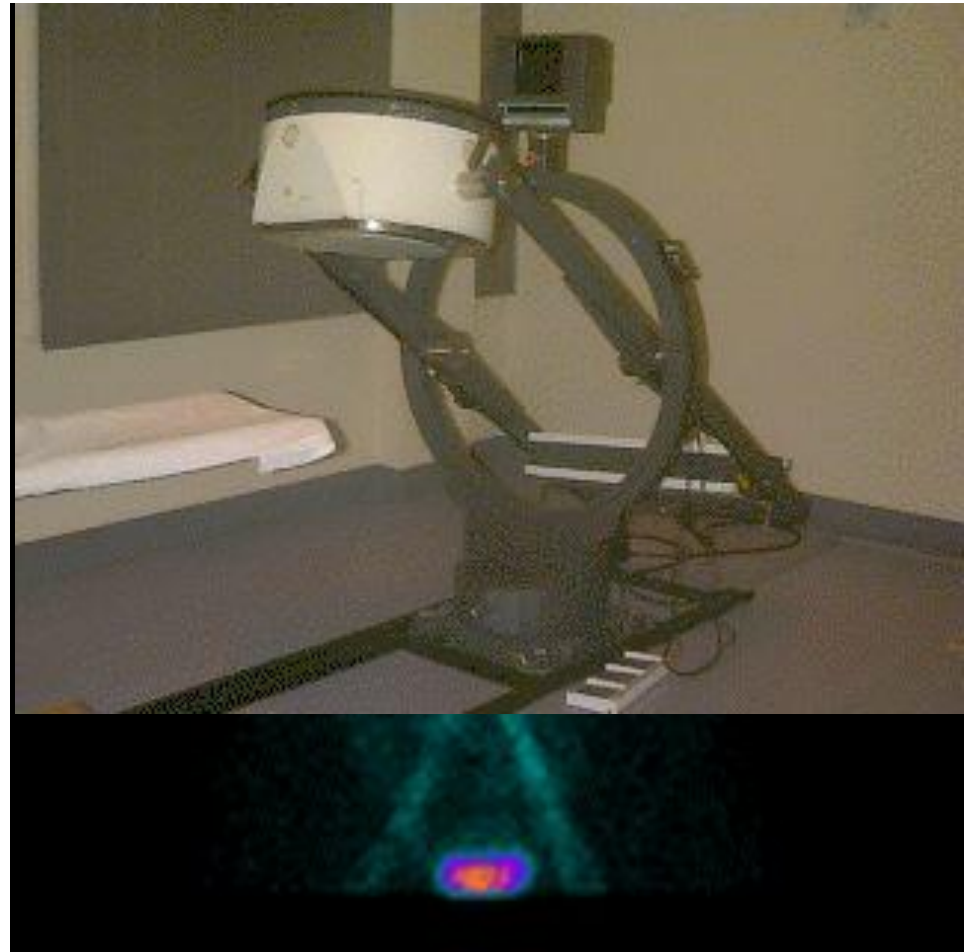
You do not need to know details of these radioactive tracers for your exam.

Gamma Cameras

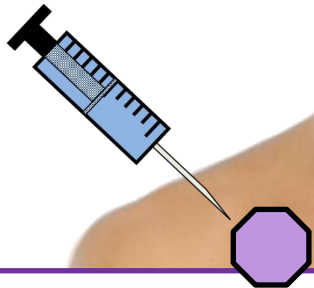
A gamma camera is often used to detect the gamma radiation emitted as the radioactive isotope in the tracer decays.

Radioactive tracers can detect gamma radiation from tracers “in real time” so doctors can watch as a substance moves through blood vessels or organs.

The image shows a radioactive tracer entering the kidneys before being excreted into the bladder.

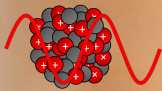


Radioactive Tracers



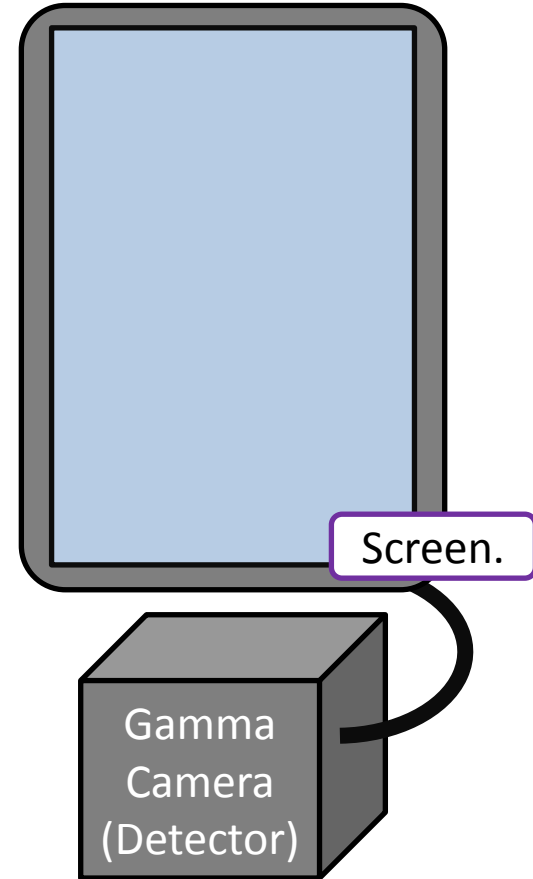
Patient injected with
(or swallows)
radioactive tracer

Tracer contains a
radioactive isotope which
decays, emitting gamma
radiation.



Specific tracers concentrate
in certain organs – this one
concentrates in the kidney.

Tracers may be transported
around the body for
example in the blood or
through the digestive
system.



Screen.

Gamma
Camera
(Detector)

Gamma radiation is
detected by the gamma
camera.

Is It Safe?

We know that radiation can be harmful – so is it safe to get someone to swallow or be injected with a radioactive isotope?

The radioactive isotopes chosen:

- Have **very short half lives** (normally hours) so the amount of radiation emitted will *rapidly fall to a safe level*.
- Are **not themselves poisonous** and do not decay into something harmful.

This means there is only a *very* small risk to the patient from the radioactive tracer. This risk is very much **outweighed** by the benefit to the patient of having an accurate diagnosis without needing surgery (which is much riskier).



TPS: Radioactive Tracers

- A patient has one blocked kidney while the other is working normally.
- They are injected with a **radioactive tracer** that will concentrate in the kidneys, before being excreted into the urine and moving down the ureter to the bladder.
- A radiation detector is placed against each kidney.

Think, Pair, Share: Answers.

1. The left kidney is blocked – the radioactive tracer is not being excreted into the urine and is staying in the left kidney.
2. Gamma radiation is used because it is the most penetrating (and so can leave the body to be detected) and least ionising.
3. An isotope with a short half life is used.

Treatment: Radiotherapy

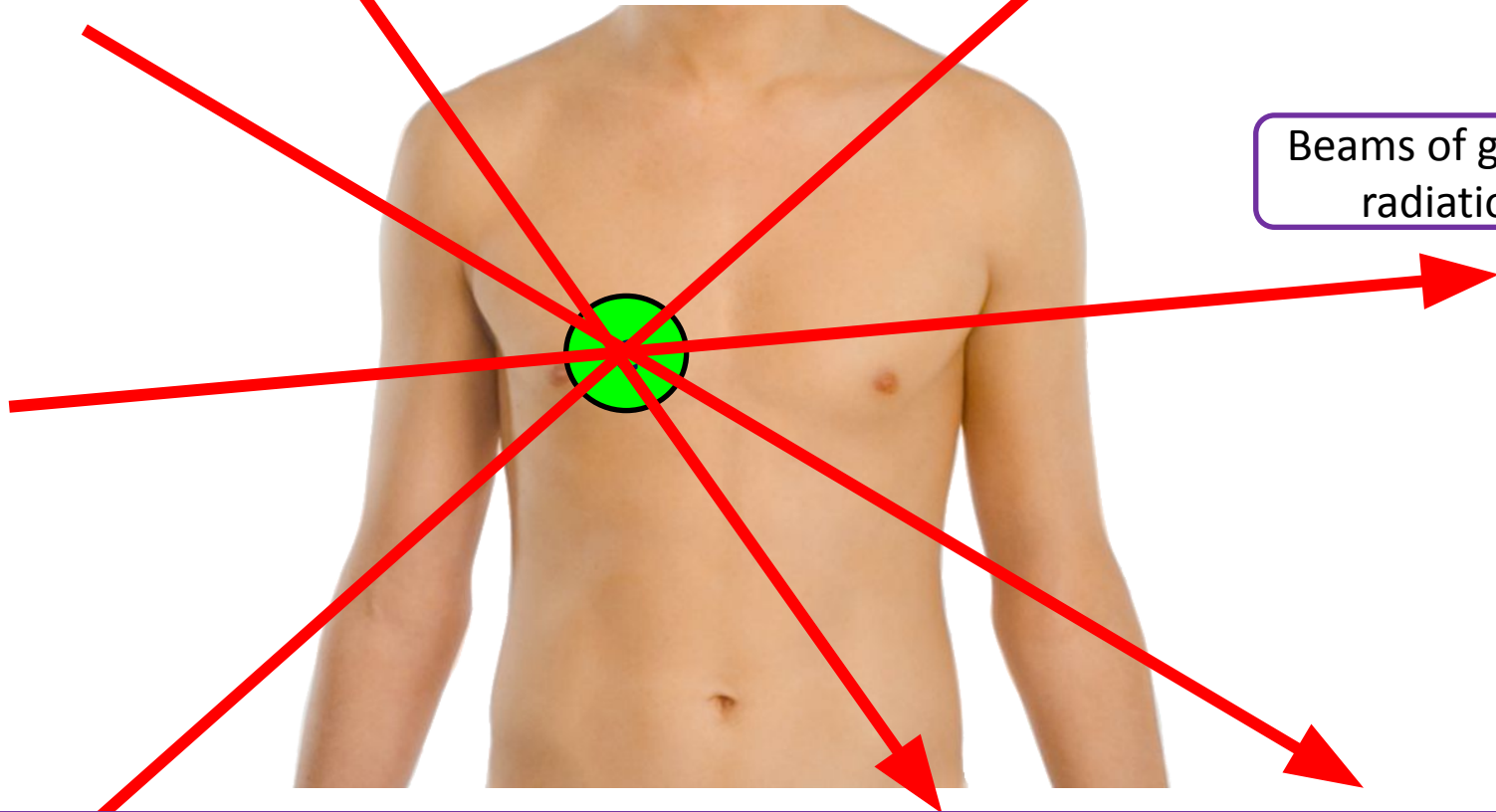
Last lesson we looked at how irradiation could be used to kill cells and tissues that are not wanted – for example bacterial cells on medical equipment or food.

Irradiation can also be used to kill cancer cells in the body.
This is called **radiotherapy**.

This is *one form* of
radiotherapy

Gamma Knife

Beams of gamma
radiation.



This patient has a cancerous tumour in his lung, shown by the green blob.
Gamma radiation is directed at the tumour from multiple directions.
This means that the cancer cells receive a high dose of radiation, but the healthy cells that are also in the path of each beam only receive a low dose.



TPS: The Gamma Knife

Using gamma rays to destroy cancer cells inside the body is called a “gamma knife”.

Just as when we irradiate food or medical instruments, the person does not become radioactive.

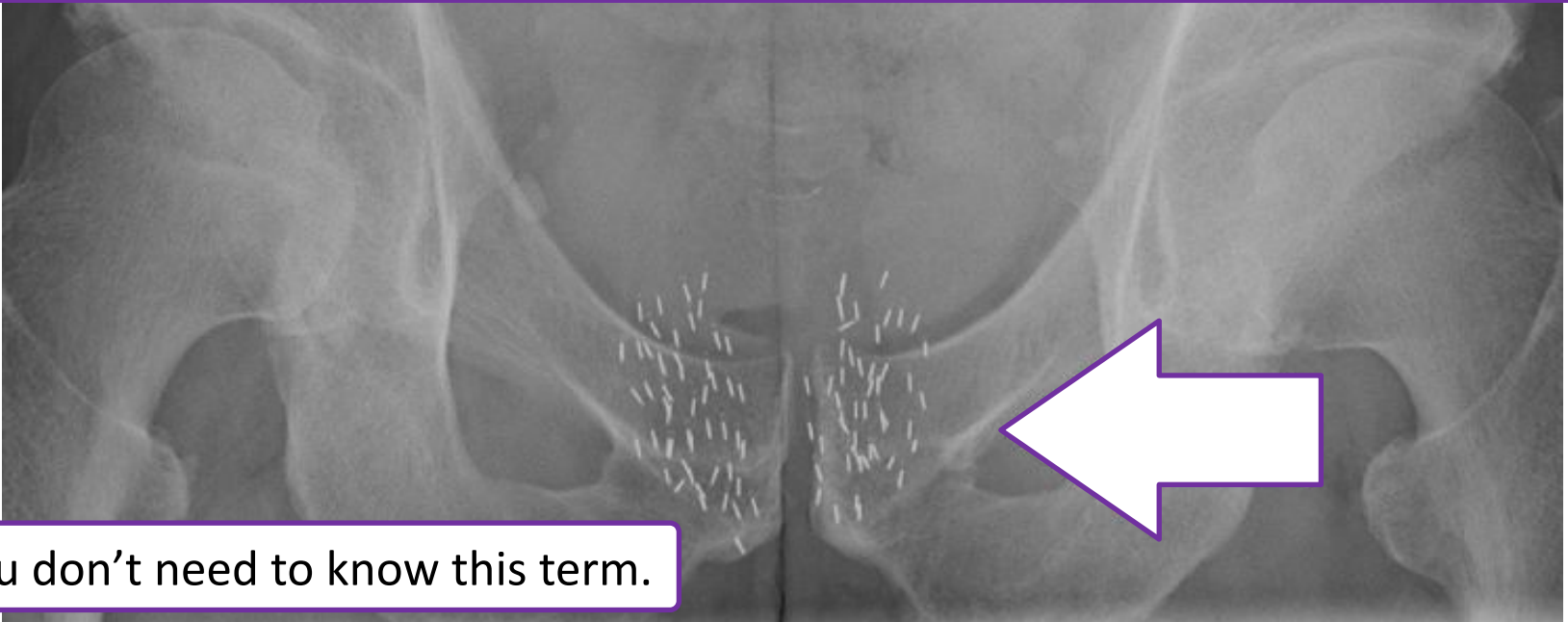
Think, Pair, Share: Answers.

1. Gamma radiation is used because it is the most penetrating, so it is able to reach cancer cells / tumours deep inside the body.
2. Unlike when using radioactive tracers, which are inserted into the patient's body, the radioactive source used for the “gamma knife” stays outside their body. Therefore we *do not* need to use a source with a short half life.

This is *another* form
of **radiotherapy**

Brachytherapy

- Another way that radiation is used to kill cancer cells is called brachytherapy*.
- Small “beads” or “rods” of a radioactive material are placed near to a tumour – on the skin for skin cancer, or *inside the body* for other cancers.
- The image below shows pellets of a radioactive isotope inserted into a man who has prostate cancer.



* You don't need to know this term.



TPS: Brachytherapy

Sometimes the implants are removed after treatment has finished, or they might remain in place.

Think, Pair, Share: Answers.

1. Sources that produce alpha and (more often) beta radiation may be used because the radioactive source is right next to the target cells and so radiation does not have to travel through the body.
2. Sources with a long half life can be used if the implants are removed, but sources with a relatively short half life will be used if they are left in place.

1 minute to discuss with the person next to you, before I choose people at random.

Learning Check: Quick Questions

Diagnosis.

...a gamma camera.

Gamma is the most penetrating (so can leave the body) and the least ionising (so will do the least damage as it does so).

Irradiation.

So that the cancer cells in the tumour receive a high dose of radiation but the healthy cells / tissues in the way only receive a low dose.

A source with a short half life will be used so that the patient is exposed to as little radiation as possible.

Radiation in Medicine

Tasks:

1. Describe how radioactive tracers are used to explore internal organs. [3 marks]
2. Which of the radioactive isotopes in the table below would be most suitable for use in a radioactive tracer? Explain your answer. [2 marks]
3. What else would we need to know about the radioactive tracer before we decided it was safe for use on a patient?
4. Describe how “gamma knife” therapy is used to destroy cancer cells inside the body. [2 marks]
5. Which of the radioactive isotopes in the table below would be most suitable for use in gamma knife therapy? Explain your answer. [2 marks]

Challenge: Suggest which of the isotopes in the table would be best for permanent insertion into a patient who had prostate cancer.

Isotope	Radiation Emitted	Half Life
Technetium-99	Gamma	6 hours
Iridium-192	Beta	73 days
Curium-242	Alpha	160 days
Cobalt-60	Gamma	5.3 years
Caesium-134	Beta	2.1 years

Answers

1. Describe how radioactive tracers are used to explore internal organs.

A tracer is swallowed by or injected into the patient (1). The tracer contains a radioactive isotope that decays and emits gamma radiation (1). The gamma rays can be detected outside of the patients body using a gamma camera / detector (1).

2. Which of the radioactive isotopes in the table below would be most suitable for use in a radioactive tracer? Explain your answer.

Technetium-99 as it is a gamma emitter (1) and has a short half life (1).

3. What else would we need to know about the radioactive tracer before we decided it was safe for use on a patient?

The tracer needs to not be poisonous / toxic itself.

Answers

4. Describe how “gamma knife” therapy is used to destroy cancer cells inside the body.

Gamma radiation is directed at the tumour from multiple directions (1). This means that the cancer cells receive a high dose of radiation, but the healthy cells that are also in the path of each beam only receive a low dose (1).

5. Which of the radioactive isotopes in the table below would be most suitable for use in gamma knife therapy? Explain your answer.

Cobalt-60 as it is a gamma emitter (1) and has a long half life so will not need to be replaced too often / will emit a fairly constant amount of gamma radiation (1).

Challenge: Suggest which of the isotopes in the table would be best for permanent insertion into a patient who had prostate cancer.

Iridium-192 as it is a beta emitter and has a half life that is neither very short nor very long.

Background Radiation

Radiation Dose: The Sievert

We have already learned that we measure the **activity** of a radioactive isotope in **becquerels (Bq)**.

However the becquerel is used to measure **activity**, which is the amount of isotopes that *decay* per second, and that *decay* may produce alpha, beta *or* gamma radiation – all of which have different effects on a person's body.

For that reason we have another measurement – the **sievert**. This measures the **dose** of radiation you receive – in other words it measures how much the radiation you receive could affect your health.

You need to be aware of sieverts but you don't need to remember the word!

This hotel in Japan is near to the Fukushima nuclear power plant where a nuclear accident occurred in 2011.

The sign shows that guests will receive a **dose** of 0.09 *micro*Sieverts (0.000 09 millisieverts) every hour. Would you believe me if I told you that was less than you get from eating one banana?



～お知らせ～
現在表示されている空間線量は、
屋内の空間線量となっております。
！注意！
壊れたり、怪我をする恐れがありますので、足元にご注意下さい。
また、精密機械の為、手を触れないようお願いいたします。

Radiation Dose: The Sievert

You need to know:

1,000 millisieverts (mSv) = 1 sievert (Sv).

8 Sv: Fatal dose, untreatable.

4 Sv: Severe radiation sickness, sometimes survivable with treatment.

2 Sv: Severe radiation sickness, sometimes fatal.

400 mSv: dose likely to cause radiation sickness if received in a short time.

20 mSv: dose that someone working with radioactive materials is allowed to receive in one year.

6 mSv: average dose from spending an hour at Chernobyl now.

0.4 mSv: dose from a mammogram (breast scan).

0.08 mSv: dose from flying from New York to London.

0.07 mSv: average dose from living in a brick building for a year.

0.0001 mSv: dose from eating one banana (which contains tiny amounts of radioactive potassium isotopes).

Background Radiation

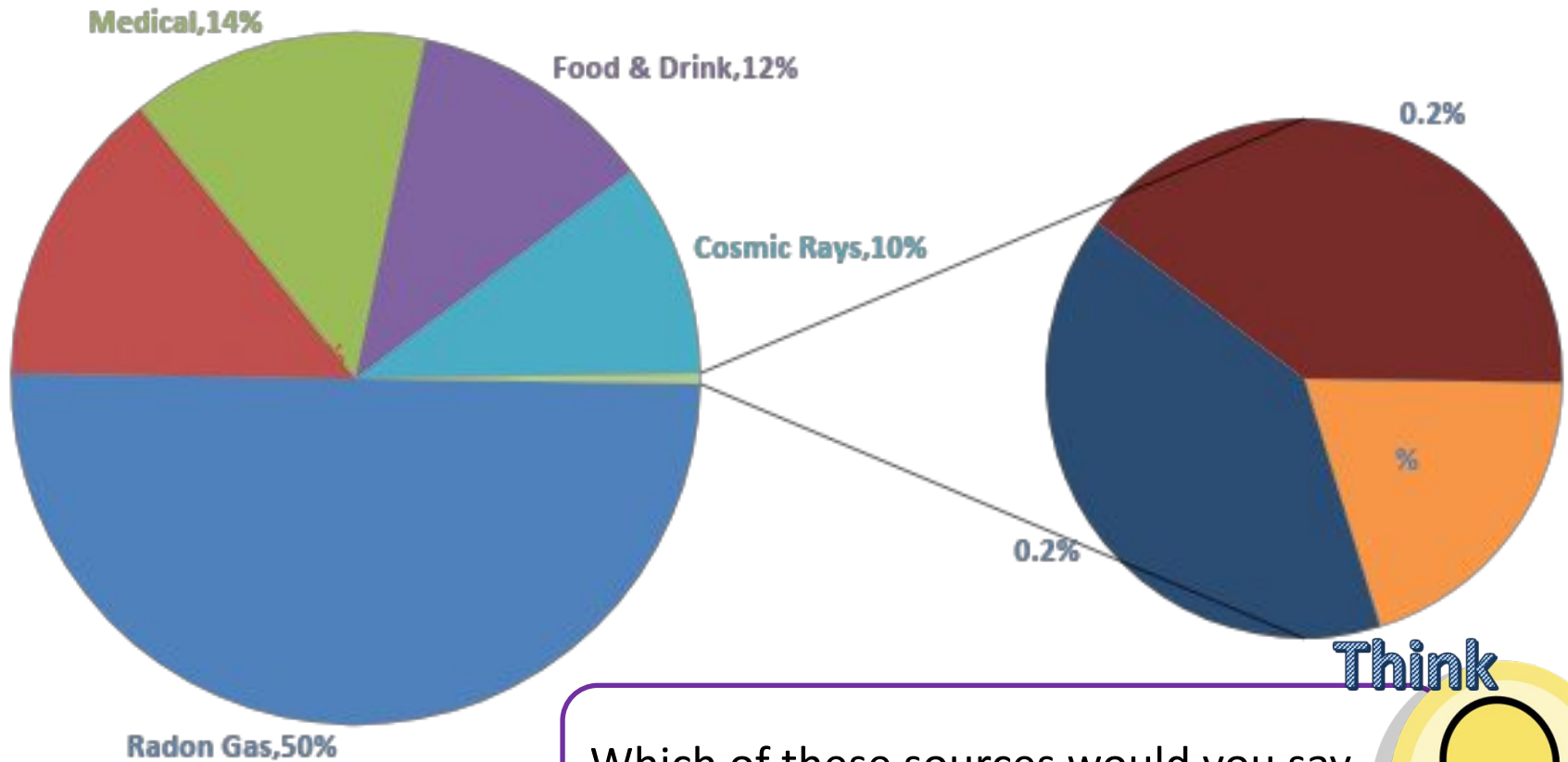
The list of radiation doses from different activities might have surprised you. In fact, **the average person in the UK receives a yearly dose of radiation of 2.7 mSv.**

This is because of **background radiation** which is the radiation that is around us all the time.

Radioactive isotopes occur naturally. Because of this we are all constantly exposed to a low level of radiation.

- Most background radiation is from natural sources.
- A small amount of the background radiation is from artificial (“man-made”) sources

Sources of Background Radiation



Which of these sources would you say are natural? Which are artificial?



Background Radiation: Natural and Artificial

Natural Sources

Radon gas – 50%
Ground and buildings – 14%
Food and Drink – 12%
Cosmic Rays – 10%

Artificial (“man-made”) Sources

Medical – 14% (*e.g. X-Rays, nuclear medicine such as we have been learning about*)
Nuclear weapons (*use and testing*) – 0.2%
Air travel – 0.2%
Nuclear power – 0.1%

Background Radiation

The level of **background radiation** that a person receives will be affected by a number of factors.

For example:

- Where you live.
- Your job.
- Particular hobbies etc.

Correcting the Count

- Scientists who want to record the activity of radioactive sources using a Geiger-Muller tube must take account of the **background radiation**.
- They record background radiation by taking several readings and averaging them.
- They then subtract this from the reading they receive when pointing the Geiger-Muller tube at the radioactive source.
- *This comes up a lot in exam questions!*

1) Average background count is $(15+22+17) \div 3 = 18$ cps

2) Therefore the corrected count is $186 - 18 = 168$ cps

Sources of Background Radiation

Radon Gas

- Uranium in rocks decays, producing other radioactive isotopes (e.g. Radon) that then decay.
- You are more likely to be exposed to a lot of background radiation from rocks if you spend time **underground** (e.g. a miner) or if you live in specific parts of the country.
- Radon diffuses into the air from rocks and soil and can build up in houses – especially if poorly ventilated.
- Radon has a half life of 3.8 days and is an alpha emitter.

As it is a gas it will be breathed in – and as we know, alpha particles are most dangerous *inside* the body as they are highly ionising (so damage cells) but weakly penetrating (so don't escape the body)

UK Radon Map

This map shows the areas of the UK where radon gas is more prevalent (due to the particular rocks in the ground)

Radon potential (1km grid)

Less than 1%



1-3%



3-5%



5-10%



10-30%



Greater than 30%



Cosmic Rays

- **Cosmic rays** are charged particles that travel at 300 000 km/h
- Cosmic rays bombard the Earth every moment of the day and night from all directions.
- The higher up you are, the more Cosmic Rays you will be exposed to.
- This means that you will receive a higher dose of background radiation, due to cosmic rays, if you:
 - live at a higher altitude
 - fly a lot



- 1) The atmosphere absorbs cosmic rays
- 2) We can tell that cosmic rays do not come from the sun because they reach all parts of the Earth both in the daytime and at night (when that side of the Earth is pointing away from the Sun).

Artificial Sources

Medicine accounts for most of the **artificial** background radiation:

- X-rays
- nuclear medicine such as we have learned about in this lesson.

However there are other sources...

Nuclear Accidents



Nuclear power, particularly **nuclear accidents** such as at Chernobyl and Fukushima can cause **contamination** that forms part of the background radiation.

In this picture, inspectors from the *International Atomic Energy Agency* inspect Fukushima – as you can see they are taking safety precautions.

Nuclear Weapons



There have been 2,056 **nuclear weapons tests** (including 45 by the UK) which contribute a small amount to the total amount of **background radiation** because of **contamination**.

Nuclear weapons have also been used twice – both times on Japan in 1945.

Radioactive fallout is radioactive material that ends up in the upper atmosphere after a nuclear blast, and then "falls out" of the sky. This can **contaminate** a large area.

1 minute to discuss with the person next to you, before I choose people at random.

Learning Check: Quick Questions

Radon gas.

Medicine.

Sieverts

130 cps.

The rocks in some parts of the country naturally release more Radon gas.

Any sensible suggestion e.g. caving, air travel.

Background Radiation and Dose

Tasks:

1. Describe what is meant by “background radiation”.
2. A student says “human activities are the cause of most background radiation”. Is this true or false? Explain your answer.
3. Describe why some people receive a higher dose of background radiation than others. Use an example to illustrate your answer. [2 marks]
4. Why are sieverts used to measure the dose of radiation a person receives and not becquerels?
5. Describe how you would account for background radiation when using a Geiger-Muller tube to measure the activity of a radioactive source. [3 marks]
6. Describe how nuclear accidents and nuclear weapons can contribute to background radiation.

Challenge: Women over 50 are invited for a mammogram (scan of the breast tissue) every 3 years, to help detect breast cancer. Each mammogram exposes them to 0.4 mSv of ionising radiation. The annual dose of radiation for someone living in the UK is 2.7 mSv.

How would you convince someone that this is an acceptable risk?

Answers

1. Describe what is meant by “background radiation”. **The radiation that is around us all the time.**
2. A student says “human activities are the cause of most background radiation”. Is this true or false? Explain your answer. **This is false, as around 85.5% of the background radiation around us is from natural sources.**
3. Describe why some people receive a higher dose of background radiation than others. **Their location, occupation and activities can affect the amount of background radiation they receive (1). For the second mark, any sensible answer e.g. taking many flights, living in a part of the country with more Radon gas (1).**
4. Why are sieverts used to measure the dose of radiation a person receives and not becquerels? **Becquerels are used to measure activity (1) but do not take into account the type of radiation, which may be more or less harmful OR sieverts take into account the health effects / harm of different types of radiation (1)**
5. Describe how you would account for background radiation when using a Geiger-Muller tube to measure the activity of a radioactive source. **Measure the background radiation (1) three times and take an average (1). Subtract this from the count (1)**

Answers

6. Describe how nuclear accidents and nuclear weapons can contribute to background radiation. **Both cause contamination (called “fallout” when it is caused by a nuclear weapon) which contributes to background radiation (1)**

Challenge: Women over 50 are invited for a mammogram (scan of the breast tissue) every 3 years, to help detect breast cancer. Each mammogram exposes them to 0.4 mSv of ionising radiation. The annual dose of radiation for someone living in the UK is 2.7 mSv.

How would you convince someone that this is an acceptable risk?

Any sensible answers for example:

- Risk from cancer is reduced.
- Treatment for cancer that is not detected early might entail greater exposure to nuclear radiation through treatment (e.g. gamma knife).
- Information to put dose in context e.g.:
 - Much less than average annual radiation dose (2.7 mSv).
 - Much less than dose that someone working with radioactive materials is allowed to receive in one year (20 mSv).
 - Equal to 5 transatlantic flights.

Exam Style Questions

Please mark this in your book as
“exam questions”

1 The average person in the UK receives a dose of radiation of 2.7 millisieverts each year, however an individual's dose can be higher or lower than this.

a) Suggest **two** reasons why the background radiation a person receives from natural sources could be higher than average.

[2 marks]

- Living in an area where there is more radon gas (1)
- Cosmic rays: living at a higher altitude (accept taking more flights) (1)

b) The majority of background radiation is from natural sources, but some is from artificial sources. Name two artificial sources of background radiation.

[2 marks]

- Nuclear Medicine / X-rays
- Nuclear accidents / nuclear power
- Nuclear weapons (tests)
- Air travel

- 1 c) A scientist used a Geiger-Muller tube to measure the radioactivity of an isotope. The counter recorded 612 counts in 60 seconds. The background radiation count rate was 0.4 counts per second.

Calculate the count rate for the sample. [3 marks]

Count rate: $612 \div 60$ (1)

Count rate = 10.2 (counts per second) (1)

Corrected count rate: $10.2 - 0.4 = 9.8$ (counts per second) (1)

2 A radioactive tracer is injected into a patient. The tracer contains the isotope technetium-99 which emits gamma radiation and has a half life of 6 hours

a) Explain why an isotope with a short half life was chosen [2 marks]

The level of radiation will decrease quickly (1)
To a low level / safe level (1)

b) Explain why an isotope that emits gamma radiation was chosen rather than one that emits alpha radiation. [2 marks]

- Gamma will escape / pass through the body (1).
- Gamma is the least / is weakly ionising (1)

Or reverse arguments

2 c) Gamma radiation can be used to kill cancer cells deep inside a patient's body. In some forms of this treatment the patient does not move while the gamma radiation source moves in a circle around them.

Explain why the source of radiation is moved around the patient rather than kept stationary. [2 marks]

- So that the cancer cells receive a full dose (or more) radiation
- But healthy cells (in the path of the radiation) only receive a low dose

Self-Evaluation

Lesson Focus and Learning Objectives

Lesson Focus:

How is radiation used in medicine?

What is background radiation?

Learning Objectives (To Be Able To...):

- Describe how nuclear radiation is used in medicine to explore internal organs and to control or destroy unwanted tissue.
- Explain why a specific radioactive source is most suitable for a given situation based on the type of radiation it emits and its half life.
- Explain why the hazards associated with a radioactive source is related to its half life.
- Describe what is meant by “background radiation” and its natural and artificial sources.
- Describe how we measure radiation dose.
- Explain why factors such as location and occupation can affect background radiation and radiation dose.

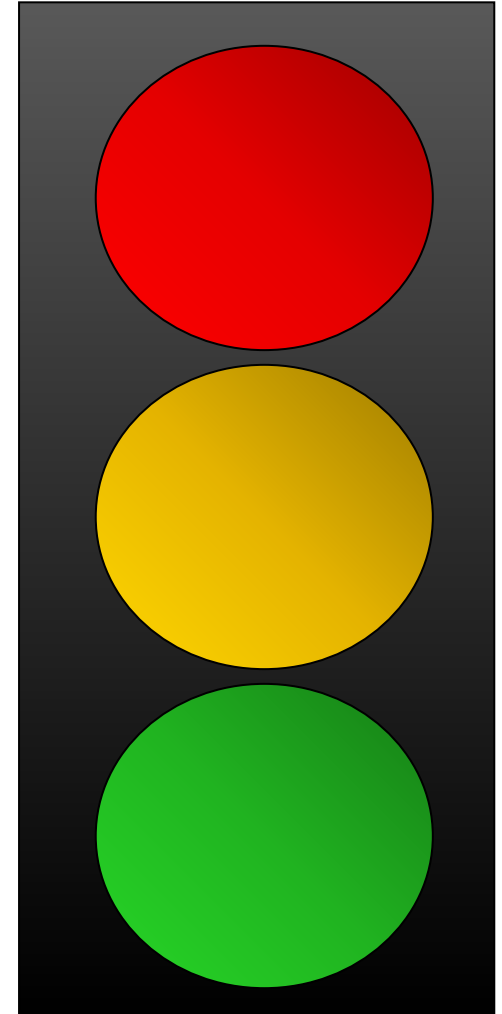
Self-Evaluation

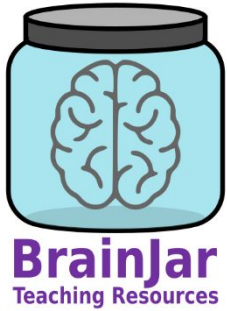
Lesson Focus: How is radiation used in medicine?
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- Describe what is meant by “background radiation” and its natural and artificial sources.
- Describe how we measure radiation dose.
- Explain why factors such as location and occupation can affect background radiation and radiation dose.

For each of the learning objectives, rate your progress towards completing them using Red, Amber or Green.

- If you are green in every area then what has helped you / what have you done to make you successful?
- If you are Amber or Red what do you need to know, do, or be helped with, in order to make you green?





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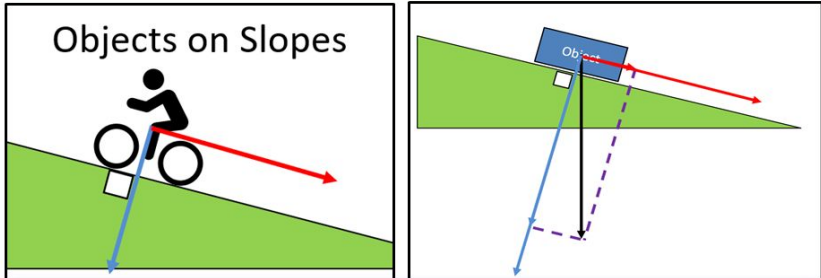
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- Ideal for non-Physicists
- Anim

- **Exam Style Questions**
- **Answers**

Example Resource

**No-Prep Complete Lesson**
GCSE Science / Physics
Resolution of Forces

Objects on Slopes



- Complete **No-Prep** Lesson
- Ideal for non-Physicists
- Anima

- **Exam Style Questions**
- **Answers**

Example Resource

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